

Program : Diploma in Instrumentation Engineering	
Course Code : 6087	Course Title: Automation Lab
Semester : 6	Credits: 2.5
Course Category: Program Core	
Periods per week : 4 (L: 0 T: 1 P: 3)	Periods per semester: 60

Course Objectives:

- To get hands-on experience with the Ladder diagram programming concepts for PLC.
- To expose students to the advanced control methods used in industries
- To prepare P&I diagrams for the simple processes
- To gain knowledge of DCS and SCADA, that are widely used in process industries

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of process control		Process control Instrumentation	4
Basic Knowledge of IT skills and basic problem solving		Introduction to IT systems	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implementation of final control elements in process control applications	9	Applying
CO2	Performance evaluation of Process stations	12	Applying
CO3	PLC programming and its applications	30	Applying
CO4	P & I Diagrams and study of SCADA and DCS	7	Applying
	Lab Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3				
CO2	3		3				
CO3	3	3	3	3			
CO4	3	3	3				2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implementation of final control elements in process control applications		
M1.1	Design and Setup ON/OFF status of a device Using Control relay	3	Applying
M1.2	To plot the characteristic curve of P/I converter.	3	Applying
M1.3	To plot the characteristic curve of I/P converter.	3	Applying
CO2	Performance evaluation of Process stations		
M2.1	Performance evaluation of Pressure Process Loop.	3	Applying
M2.2	Performance evaluation of flow Process Loop.	3	Applying
M2.3	Performance evaluation of Level Process Loop.	3	Applying
M2.4	Performance evaluation of Temperature ProcessLoop.	3	Applying
	Lab Test - I	1	
CO3	PLC programming and its applications		
M3.1	Construct PLC ladder diagram programming for Digital Logic gate function and simulate it.	3	Applying

M3.2	Construct PLC ladder diagram programming for ON/OFF status of a device remotely operating device and simulate	3	Applying
M3.3	To know the concept of interlock for safety in process control application using PLC	3	Applying
M3.4	Construct PLC ladder diagram Programming for auto operation of 2 devices using timer and simulate it.	3	Applying
M3.5	Construct PLC ladder diagram Programming for counter and comparator and simulate it.	3	Applying
M3.6	Construct PLC Ladder Diagram Programming for a LIFT operation	3	Applying
M3.7	Water Level control using PLC.	3	Applying
M3.8	Speed control of a motor using PLC.	3	Applying
M3.9	Traffic Control using PLC.	3	Applying
M3.10	Bottle filling system using PLC.	3	Applying
CO4	P & I Diagrams and study of SCADA and DCS		
M4.1	Study of SCADA package.	2	Understanding
M4.2	Study the features of DCS.	2	Understanding
M4.3	Prepare the Instrumentation documents of given P&I Diagram.	3	Applying
	Lab Test - II	1	

Text /Reference:

T/R	Book Title/Author
R1	Curtis .D. Johnson , Process control instrumentation technology , PHI
R2	John webb, Programmable Logic Controllers - Macmillan Publishing
R3	Liptak , Handbook of Process Control

Online Resources:

Sl.No	Website Link
1	http://www.ieec.uned.es/investigacion/Dipseil/PAC/archivos/introtoplcs_SUPER.pdf
2	https://www.nfiautomation.org/FREE_Download/Technical%20Documents/PLC/%28eBook%29%20%20PLC%20Programming%20Methods.pdf
3	https://www.academia.edu/36448361/Beginners_Guide_to_PLC_Programming_How_to_Program_a_PLC_Programmable_Logic_Controller
4	http://ee.ump.edu.my/hazlina/teaching_PLC/teaching_PLC_chap3.pdf